

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.3: Temperature and Thermal Expansion — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.3: Temperature and Thermal Expansion
Driving Question	DRIVING QUESTION: Why did the sealed water bottle crack in the freezer — and what does this reveal about how all matter responds to changes in temperature?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Encounter and Predict: The Cracked Frozen Bottle	A pre-prepared sealed glass bottle filled with water that cracked overnight in a freezer, displaying a protruding ice plug and a distinctive crack-line pattern along the glass walls. Learners also examined a labelled A3 cross-section diagram showing liquid water molecules, the hexagonal ice crystal lattice, the direction of outward pressure, and the resulting crack pattern.	Water behaves anomalously when it freezes — unlike most substances, it expands rather than contracts as it solidifies. This expansion generates sufficient outward pressure to crack a sealed glass bottle. At the particle level, ice molecules in a hexagonal lattice are spaced further apart than liquid water molecules, meaning ice occupies a greater volume than the same mass of liquid water.	The cracked bottle is explained by the anomalous expansion of water: as water freezes at 0°C, its molecules arrange into a hexagonal ice crystal lattice with larger intermolecular spacing than liquid water. Because the bottle was completely sealed with no room for expansion, the increasing volume of the freezing water generated enormous outward pressure on the glass walls, eventually exceeding the tensile strength of the glass and causing it to crack. This behaviour is anomalous because almost all other substances contract when they cool and solidify.
Lesson 2 Temperature Scales and Thermometers: Measuring the "How Hot"	At Station A, learners observed the red alcohol column in the liquid-in-glass thermometer rising when placed in warm water and falling in ice-cold water. At Station B, learners identified the constriction in the clinical thermometer that prevents the column from falling before a reading is taken. At Station C, learners observed the bimetallic strip bending toward the copper side when heated, indicating that brass expands more than copper under the same	Temperature is a measure of the average kinetic energy of particles in a substance. A thermometric property is any physical property that changes continuously, noticeably, and reproducibly with temperature. Temperature in Kelvin is calculated as $T(K) = T(^{\circ}C) + 273$. Absolute zero (0 K / -273 °C) is the theoretical point at which particle motion is at its minimum. The freezer temperature (-18 °C / 255 K) caused water molecules to slow down and	The water in the sealed bottle was at approximately room temperature (~18 °C / 291 K) when placed in the freezer. As the freezer cooled the water to 0 °C (273 K) it froze: the water molecules lost kinetic energy, slowed down, and reorganised into an ordered hexagonal lattice structure. This lattice has greater intermolecular spacing than liquid water, so the ice occupies a larger volume. Because the bottle was sealed and could not expand, the growing

	temperature change.	arrange into a hexagonal ice crystal lattice — a lattice that occupies more volume than liquid water at the same mass.	ice exerted outward pressure on the glass walls — eventually exceeding the tensile strength of the glass and causing the bottle to crack.
Lesson 3 Linear Expansion of Solids — The Stretching Rod	Learners heated a metal rod using a Bunsen burner and measured its length change using a linear-expansion apparatus with a dial gauge, recording temperature and length change data at 1-minute intervals. They also viewed a video clip of thermal expansion gaps on the SGR railway between Mombasa and Nairobi.	Learners derived the coefficient of linear expansivity (α) from their own data as the constant ratio $\Delta L / (L_0 \Delta T)$, and applied the formula $\Delta L = \alpha L_0 \Delta T$ to calculate expansion in SGR steel rails ($\Delta L \approx 12.4$ mm per 25 m rail over a 45°C rise) and Kenya Power aluminium overhead cables near Kisumu ($\Delta L \approx 25.3$ cm per 200 m cable over a 55°C rise). They updated their particle model to show that heating increases vibration amplitude and average particle separation.	When a solid is heated, its particles gain kinetic energy and vibrate with greater amplitude, increasing the average distance between neighbouring particles. This microscopic change produces a measurable macroscopic increase in length described by $\Delta L = \alpha L_0 \Delta T$. The same principle — expansion generating force against a rigid boundary — explains why the sealed glass bottle cracked when the water inside froze and expanded, and why SGR engineers must leave deliberate gaps between rails to prevent track buckling.
Lesson 4 Superficial and Volumetric Expansion — From Length to Volume	Learners set up a flask-and-capillary-tube apparatus, cooled water from room temperature toward 0°C while measuring the capillary column height at regular temperature intervals, and plotted a temperature–volume graph. They observed that the capillary column — and therefore the water volume — decreased as temperature dropped from 20°C to 4°C , then increased again as temperature dropped further from 4°C to 0°C , forming a distinctive U-shaped curve with a minimum at approximately 4°C .	Learners learned that: (1) superficial expansivity $\beta \approx 2\alpha$ and volumetric expansivity $\gamma \approx 3\alpha$ can be derived mathematically from linear expansivity α , and used in the formula $\Delta V = \gamma V_0 \Delta T$; (2) most substances contract when cooled, but water is anomalous — it expands as it cools below 4°C , and on freezing at 0°C water molecules arrange into a hexagonal ice crystal lattice with greater intermolecular spacing than liquid water; (3) water reaches its maximum density at 4°C ; (4) this anomalous expansion is the direct cause of the cracked frozen bottle in the anchoring phenomenon, as the expanding ice generated pressure exceeding the tensile strength of the glass.	The sealed glass bottle cracked because water, unlike most substances, expands when it freezes. As the water cooled below 4°C it began to expand, and on freezing at 0°C its molecules arranged into a hexagonal ice crystal lattice in which the intermolecular spacing is greater than in liquid water — so the same mass of water occupies a larger volume as ice. Confined inside a sealed bottle with no room to expand, the growing ice exerted outward pressure on the glass walls. When this pressure exceeded the tensile strength of the glass, the bottle cracked. Ordinary volume changes within one phase can be quantified using $\Delta V = \gamma V_0 \Delta T$, but the expansion of about 9% on freezing is a phase-change effect that this formula does not describe; the temperature–volume graph of water provides direct experimental evidence of this anomalous behaviour.
Lesson 5 Applications of Thermal Expansion: Engineering Solutions	Learners observed four real Kenyan applications of thermal expansion through application cards: expansion joints on the Thika Superhighway, bimetallic strip thermostats in electric irons, metal tyre fitting on wooden cart wheels, and riveted corrugated roofing. They also observed a	Learners learned that engineers use the linear expansion formula $\Delta L = \alpha L_0 \Delta T$ to calculate the exact gap size required in structures such as bridges and roads, preventing destructive buckling or cracking. They learned that bimetallic strips exploit differential expansion of two	Learners explained that the same physical principle that caused the bottle to crack — materials expanding when heated and generating force when confined — is the principle engineers consciously calculate and design around in every structure exposed to temperature changes in Kenya.

	worked calculation showing that a 10 m steel beam in Kisumu requires at least 1.65 mm of expansion gap per end to safely accommodate a 15 degree C seasonal temperature rise.	bonded metals to create a temperature-controlled switch, and that thermal expansion can be deliberately engineered to be useful (tyre fitting, thermostat) rather than only destructive (cracked bottle). They practised applying the formula to a new context involving Kenya Power copper cables expanding by 8.5 cm over 200 m on a hot Nairobi afternoon.	The cracked bottle represents an uncontrolled application of thermal expansion; Kenyan engineering infrastructure represents the controlled and calculated application of the same phenomenon, making the bottle not just a curiosity but a model for understanding why expansion joints, sagging cables, and bimetallic switches exist.
Lesson 6 Model Revision, Synthesis, and DQB Resolution — Explaining the Crack	Learners conducted a six-station Evidence Gallery Walk, revisiting and recording key findings from all five prior lessons — including particle diagrams of liquid water vs. ice, the density-vs-temperature graph showing the 4°C anomaly, and the volumetric expansion formula $\Delta V = \gamma V_0 \Delta T$ applied to a 500 ml water bottle. Learners physically placed their Lesson 1 prediction cards beside their final L6 particle models, making conceptual growth visible. The Driving Question Board was returned to the class with all accumulated sticky-note questions, and every question received either a green resolved note or an orange frontier-science note.	Learners demonstrated that water's anomalous expansion upon freezing is caused by hydrogen bonding, which arranges water molecules into a hexagonal lattice with greater intermolecular spacing than liquid water — reducing density from approximately 1.00 g/cm ³ (liquid at 4°C) to 0.917 g/cm ³ (ice at 0°C). This ~9% volume increase in a sealed, rigid glass bottle generates internal pressure exceeding the tensile strength of glass (33–100 MPa), causing the characteristic crack pattern. Learners calculated that 500 ml of water freezing expands by about 9%, $\Delta V \approx 45$ ml (a phase-change effect beyond $\Delta V = \gamma V_0 \Delta T$). They connected this knowledge to Kenyan engineering contexts: burst water pipes in the Kericho and Nyandarua highlands, expansion joints on the SGR Nairobi–Mombasa railway, and storage tank design at the Mombasa port.	The sealed water bottle cracked because water is one of the very few substances on Earth that expands — rather than contracts — when it freezes. As the temperature dropped from 4°C (the point of maximum density) to 0°C, the liquid water expanded slightly; then, on freezing at 0°C, water molecules formed hydrogen bonds that locked them into a rigid hexagonal lattice with greater spacing between molecules. This expansion increased the volume of water by approximately 9%, generating a pressure force directed outward against the rigid glass walls. Because the glass could not stretch to accommodate this expansion, the stress exceeded the glass's tensile strength and the bottle fractured along its weakest lines. Ordinary thermal expansion within one phase — quantified by the volumetric expansion formula $\Delta V = \gamma V_0 \Delta T$ — explains why engineers build expansion gaps into railway tracks, design flexible couplings in water pipelines, and specify temperature-rated containers for cold storage across Kenya and the world.

ATTRIBUTION AND LICENSING

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